Number: 01
Tittle: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields
Link: https://arxiv.org/pdf/1611.08050.pdf
Date: 01/22/2023
Datasets: COCO 2016 keypoints challenge & MPII human multi person

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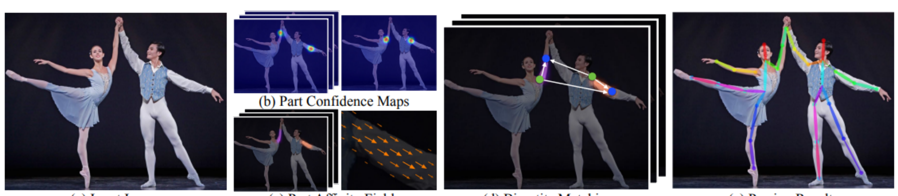
Allows the first real-time system for multi-person 2D pose detection by implementing the first bottom-up approach that is able to create association between body parts with the use of Part Affinity Fields (PAF).

Method:

- Input: a color image of size w × h.
- Feedforward network simultaneously predicts: Confidence maps & Affinity fields. This is made by a multistage iterative prediction architecture.
- Finally, the confidence maps and the affinity fields are parsed by greedy inference to output the 2D key points for all people in the image.
- Output, the 2D locations of anatomical key points for each person in the image.



(a) Input Image (c) Part Affinity Fields (d) Bipartite Matching (e) Parsing Results

Part Affinity Fields:
New feature representation that preserves both location and orientation information. For each pixel in the area belonging to a particular limb, a 2D vector encodes the direction (unit vector) that points from one part of the limb to the other. Otherwise the value is zero.

Groundtruth PAF is the average PAFs of all people in the image.

During testing, we measure association between candidate part detections by computing the line integral over the corresponding PAF, along the line segment connecting the candidate part locations.

We perform non-maximum suppression on the detection confidence maps to obtain a discrete set of part candidate locations. For each part, we may have several candidates. We score each candidate limb using the line integral computation on the PAF.

Evaluation:

- MPII: mAP with PCKh threshold & Average inference/optimization time per image
- COCO: Mean average precision (AP) over 10 OKS (Object key position similarity) thresholds, AP^{50}, AP^{75}, AP^{M} & AP^{L}